

SCSIT

IV Semester

Databases

Exc. 2

Entity keys

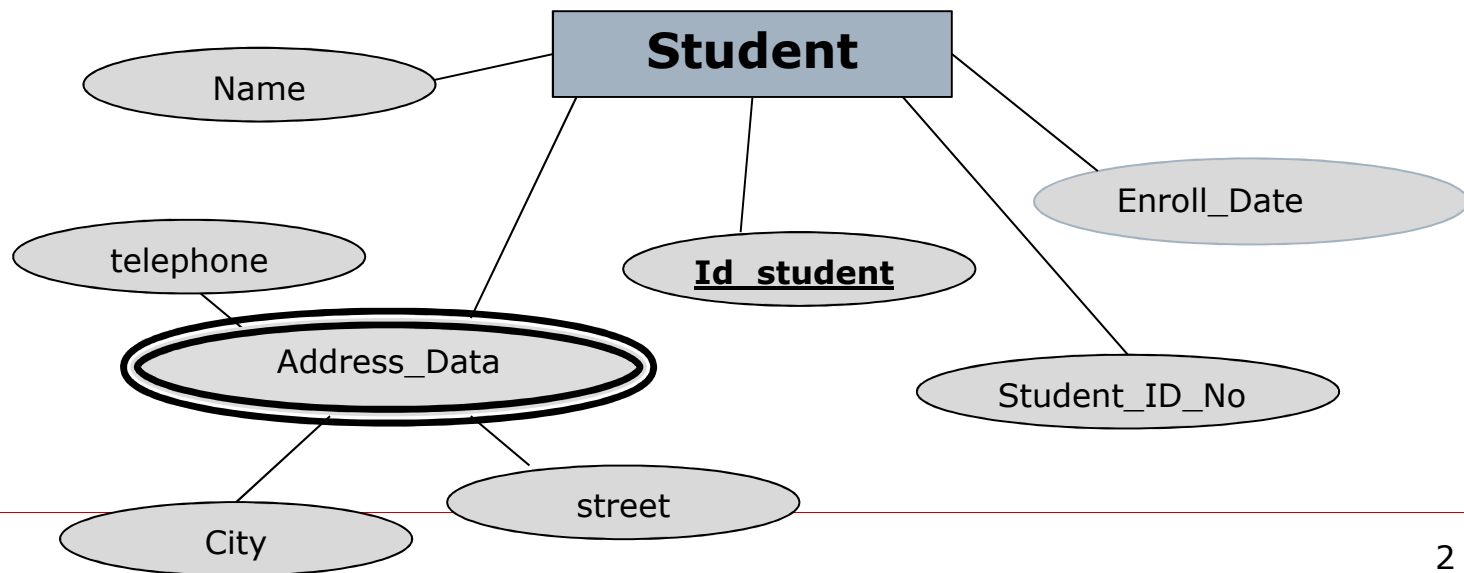
Relations (Cardinality constraints)

From last time

□ ENTITY

INSTANCE (?????)

- Attributes
- Composite attributes
- Attribute identifier



Identifier

- ❑ One or more entity attributes that can uniquely identify every instance of a given entity or relation

Identifier = KEY

- ❑ If you can not find proper attribute for this manner from the existing ones, you can define one, which will have thus role
- ❑ *Very rarely, but it is possible to find entity or relation without Key*

Example

□ Student (id_student, name, study_year)

1248	Marija Vasileva	2
1297	Martin Simonovski	2
1328	Petar Blazevski	1
1165	Slobodan jovanovski	3
1215	Aleksandar Stankov	2
1306	Trajce Ristov	1
1323	Blagoja Naskov	1
1342	Aleksandar Stefanoski	1
1363	Kosta Ilievski	1
1234	Dusan jovanovksi	3
1182	Simot Milosevik	3
1191	Petar Dimov	3
1206	Petar Stojkovski	2
1299	Gjorgji Ilievski	2
1359	Dejan Stankovik	3

In this case, **id_student** is an attribute – **entity key**, because for every entity instance, it has different value

Ass. 1

☐ Entities

- *Client* (client_no, name, PID, telephone)
- *Account* (account_no, type)
- *Add_money* (amount, transaction_date)

☐ Define keys for the given entities.

☐ Which attribute is key for add_money?

Entity relation

□ Relation (**relationship**) is connection (interaction) between several entities (one, two or more)

■ *It can have its own attributes*

□ Ex.

Pero	add_money	203043346745
<i>(client)</i>	<i>(relation)</i>	<i>(account)</i>

Types of relations

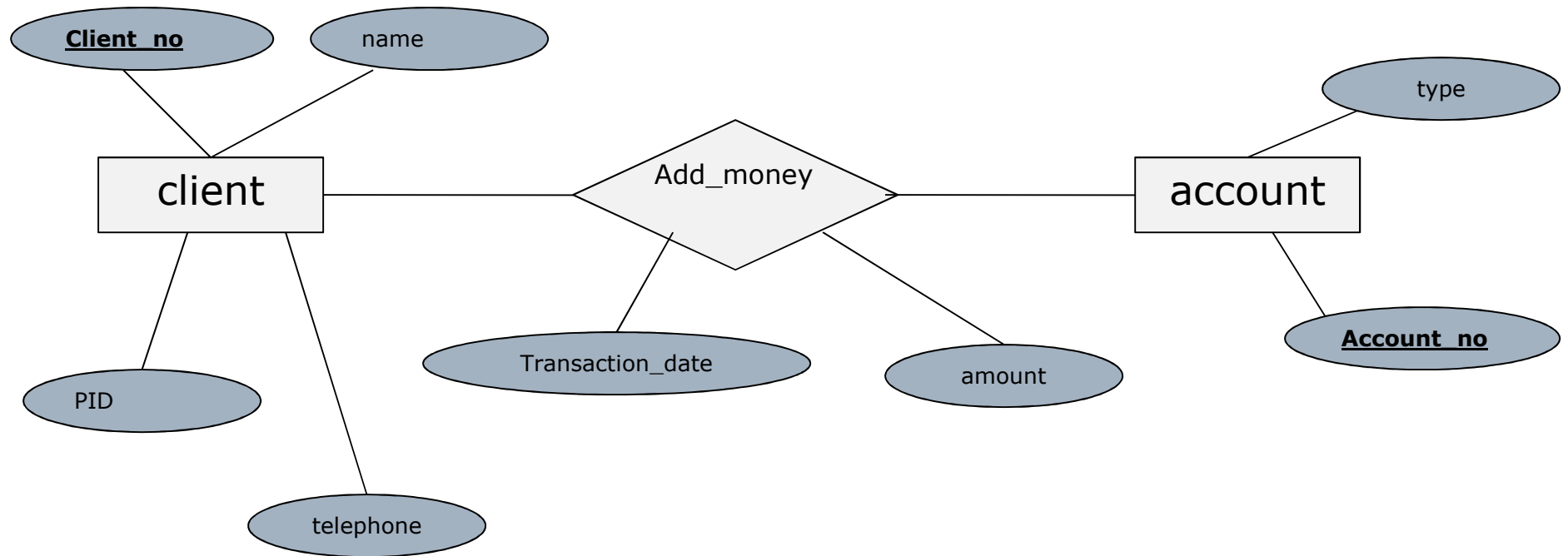
- ☐ Binary: two entities relation
- ☐ n-ary: N entities relation

What kind of relation is the previous one?

Relations in ER Diagrams

- Deltoids (diamonds) represent relation sets ($\sim$ relations)
- **Lines** connect entities with relations

ER Diagram for the example



Ass. 2

- Create ER Diagrams for the following cases:
 - Student enrolls in study year
 - Student goes to exam for some course
 - The course is part of some study year

E-R model semantics

- ☐ Primary Key
 - **One of the entities identifiers.**
 - Every entity has its own identifier (recommended)
- ☐ **Cardinality constraints**
- ☐ **Types of relations**
- ☐ **Relation degree**
- ☐ **Existence -participation constraint**

Primary key

In our example

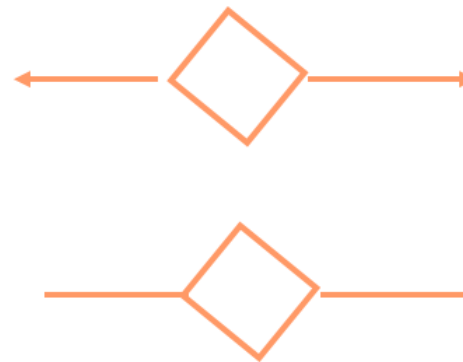
client – add_money – account

Primary keys are:

- client: client_id
- account: account_id
- add_money: (?).....
 - combination of (client_id, account_id) – deductive primary key
 - or you can add additional attribute id_add_money, which will be its primary key

Cardinality constraints

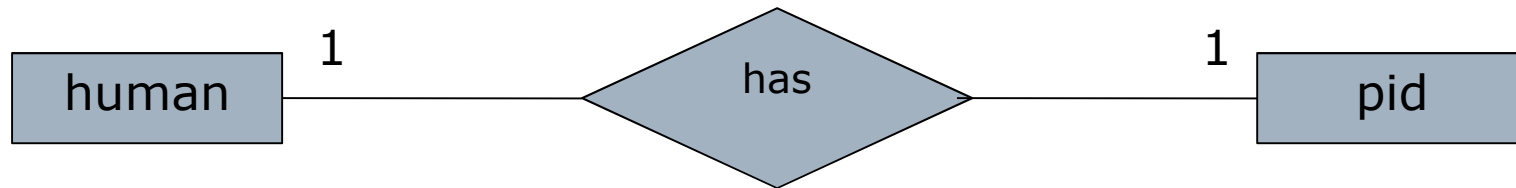
- ❑ **Number** of instances from one entity that can be in relation with one instance from another entity
- ❑ Used in binary relations
- ❑ Several types:
 - **One-to-one**
 - **One-to-many**
 - **Many-to-one**
 - **Many-to-many**



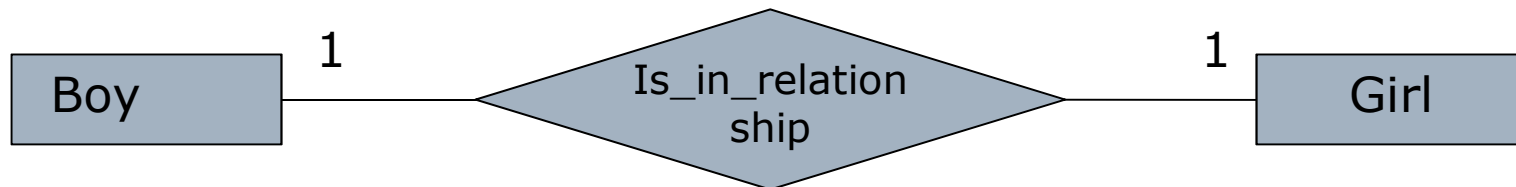
One-to-one

- Every instance from E1 entity can be in relation with exactly one instance from the E2 instance, v.v.

Ex. Every human can have only one PID (if PID is entity) and v.v.



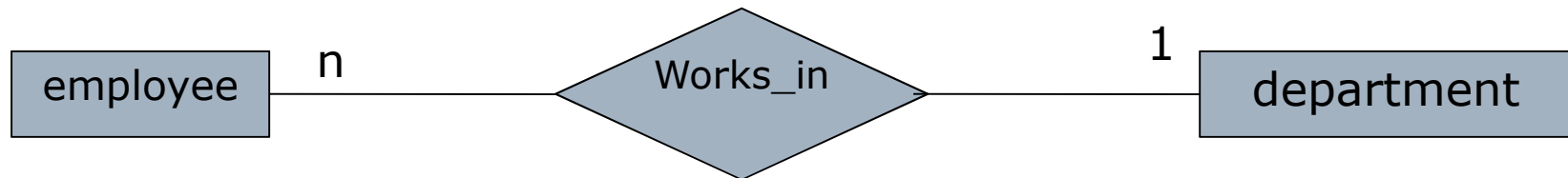
One boy is in relationship with one girl



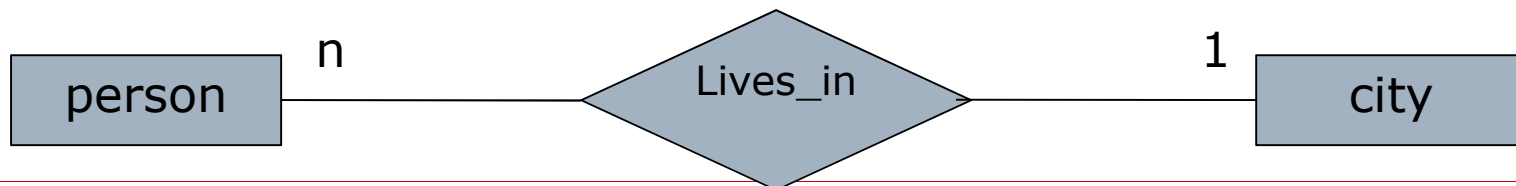
One - to - many

□ Every instance from E1 entity can be in relation with none or more instances from the E2 instance, but every instance from E2 must be in a relation with only one instance from E1.

Ex. In **ONE** department, there are **MANY** employees, but **One** employee works in **ONLY ONE** department



In **ONE** city there are **MANY** people that live, but one **PERSON** live in only **ONE** city

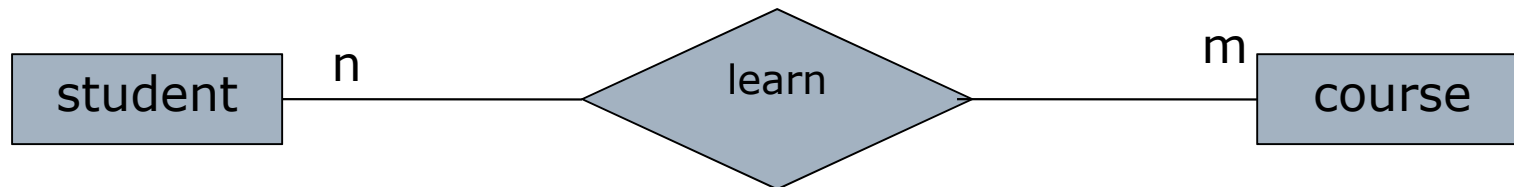


When you create DB model from this type of ER diagram, tables that represent CITY and DEPARTMENT entities are known as NOMENCLATURES.

Many-to-many

□ Every instance from E1 entity can be in relation with none or more instances from the E2 instance, and every instance from E2 can be in a relation with none ore more instances from E1.

Пр. **MANY** different students learn **MANY** different courses



When you create DB model from this type of ER diagram, you must create₁₆ so called, **intersection tables** that represent this relation (learn)

Special types of constraints

- ❑ Ex. One student can enroll at least 1 but, max. 5 course in one semester.
- ❑ One course can be enrolled by none or many (n) students.

